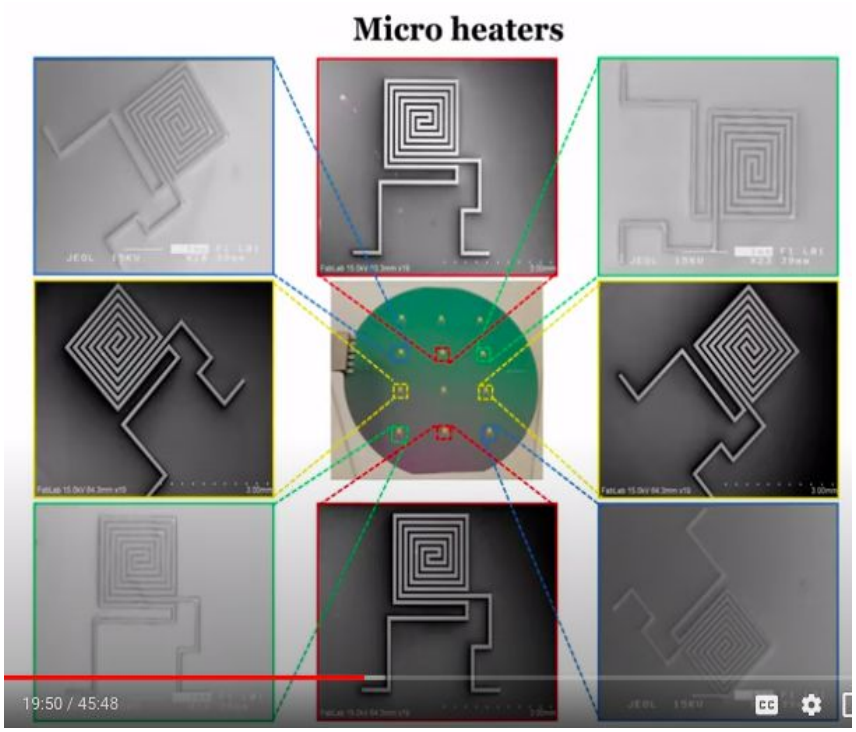
Basics of Energy Transformation: Transducers, Sensors and Actuators

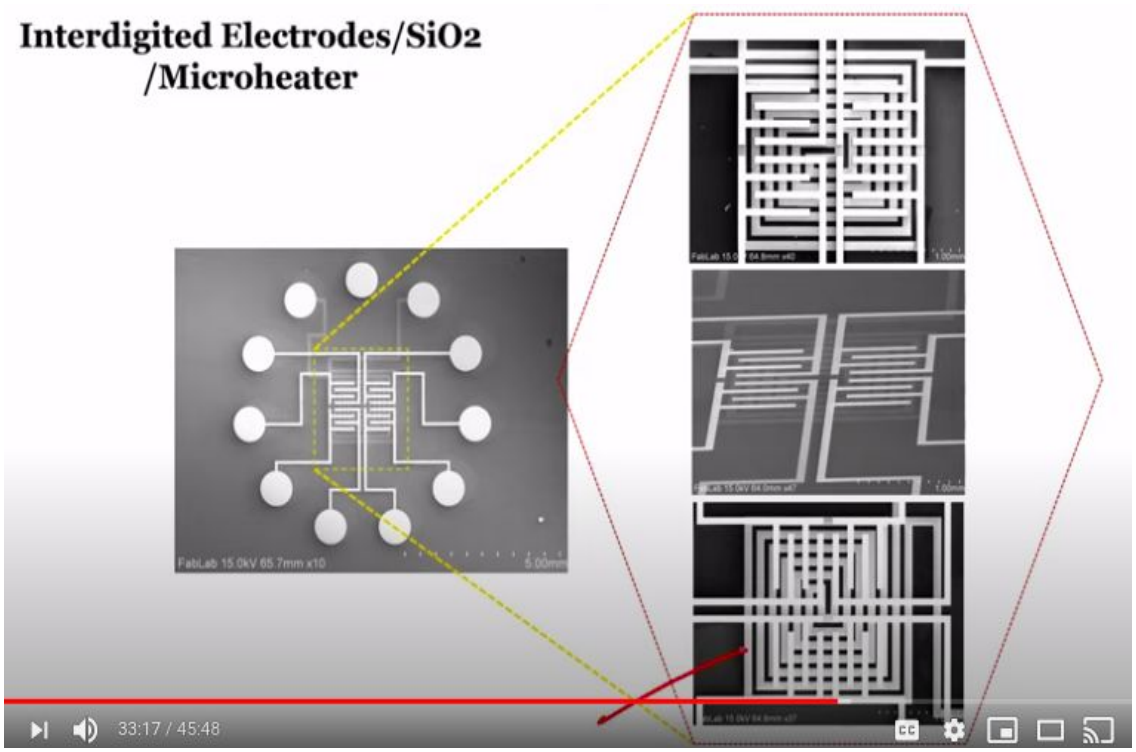
Transducer : is a device which converts one form of energy into another form of energy like thermocouple, diaphragm, LVDT etc. the transducer is the primary element and plays an important role in measurement instrumentation.

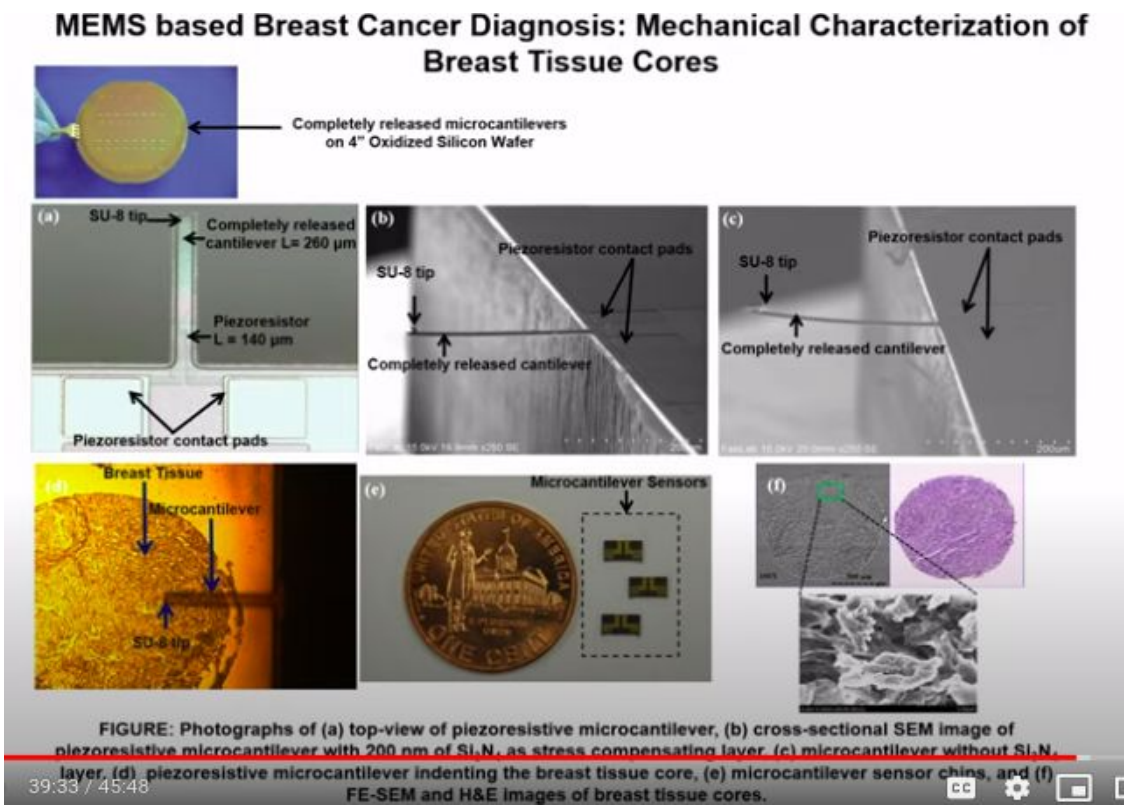
Sensor: is a device which can detect physical phenomenon like temperature, pressure, flow, level etc.

sensor=transducer + signal conditioning circuit

Note: ***that a transducer and a sensor are not the same. All sensors consist of a transducer but all transducers need not to be a sensor.***




Interdigited Electrodes/SiO2
/Microheater
33:17 / 45:48


MEMS based Breast Cancer Diagnosis: Mechanical Characterization of
Breast Tissue Cores
Completely released microcantilevers
on 4" Oxidized Silicon Wafer
(a)
SU-8 tip
Completely released
cantilever L= 260 µm
Piezoresistor
L = 140 µm
Piezoresistor contact pads
(b)
Piezoresistor contact pads
SU-8 tip
Completely released cantilever
(c)
Piezoresistor contact pads
SU-8 tip
Completely released cantilever
(d)
Breast Tissue
Microcantilever
SU-8 tip
(e)
Microcantilever Sensors
(f)
FIGURE: Photographs of (a) top-view of piezoresistive microcantilever, (b) cross-sectional SEM image of
piezoresistive microcantilever with 200 nm of Si3N4 as stress compensating layer, (c) microcantilever without Si3N4
layer, (d) piezoresistive microcantilever indenting the breast tissue core, (e) microcantilever sensor chips, and (f)
FE-SEM and H&E images of breast tissue cores.
39:33 / 45:48

MEMS based Breast Cancer Diagnosis: Electrical Characterization of Breast Tissue Cores

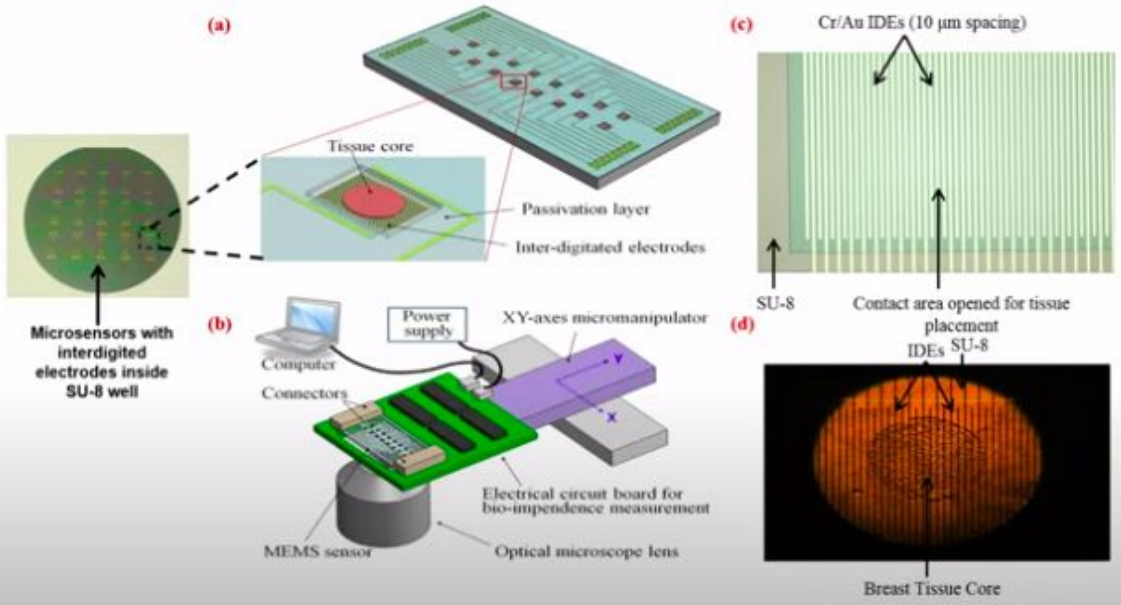
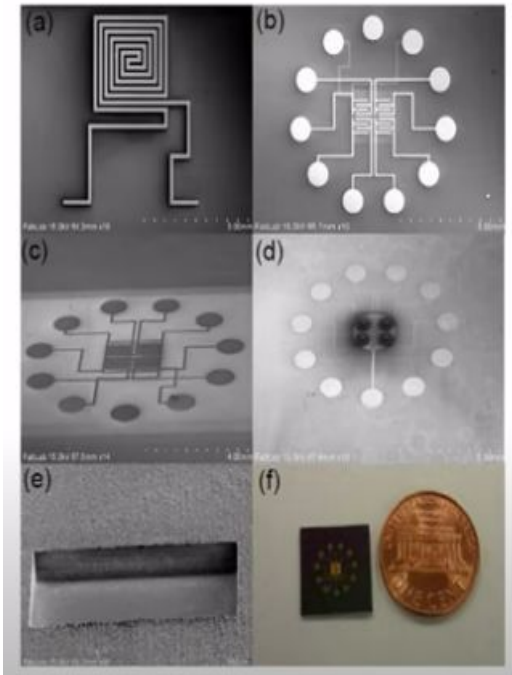


FIGURE: Schematics and optical photographs of (a) MEMS-based bioimpedance sensor for tissue microarray, (b) the semi-automated bioimpedance measurement system, (c) microchip with IDEs having 10 um width and 10 um spacing, and (d) cancerous breast tissue placed on interdigited electrodes.

15:00 / 31:27

Cancer Diagnosis Tool



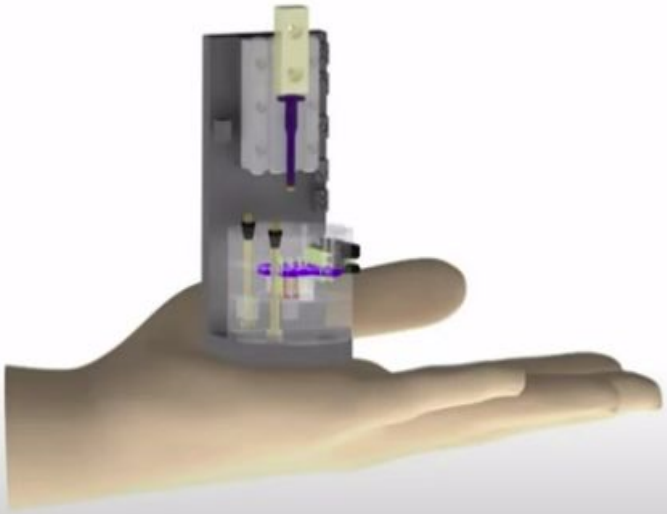


Figure. SEM images of (a) Microheater, (b) Cr/Au interdigited electrodes over microheater, (c) Sensing layer, (d) Gold coated SU-8 pillars over Cr/Au electrode over silicon dioxide over sensing layer,(e) Backside of silicon diaphragm. (f) A photograph of the biochip.

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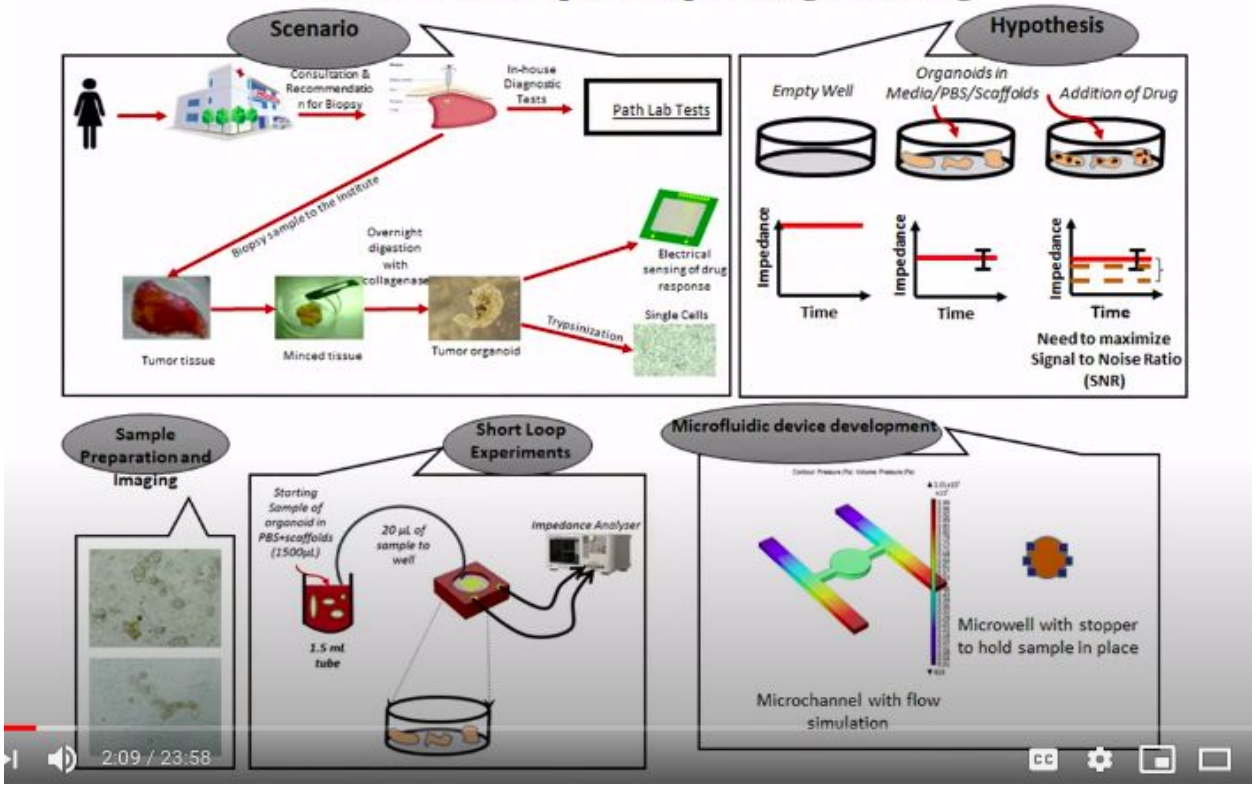
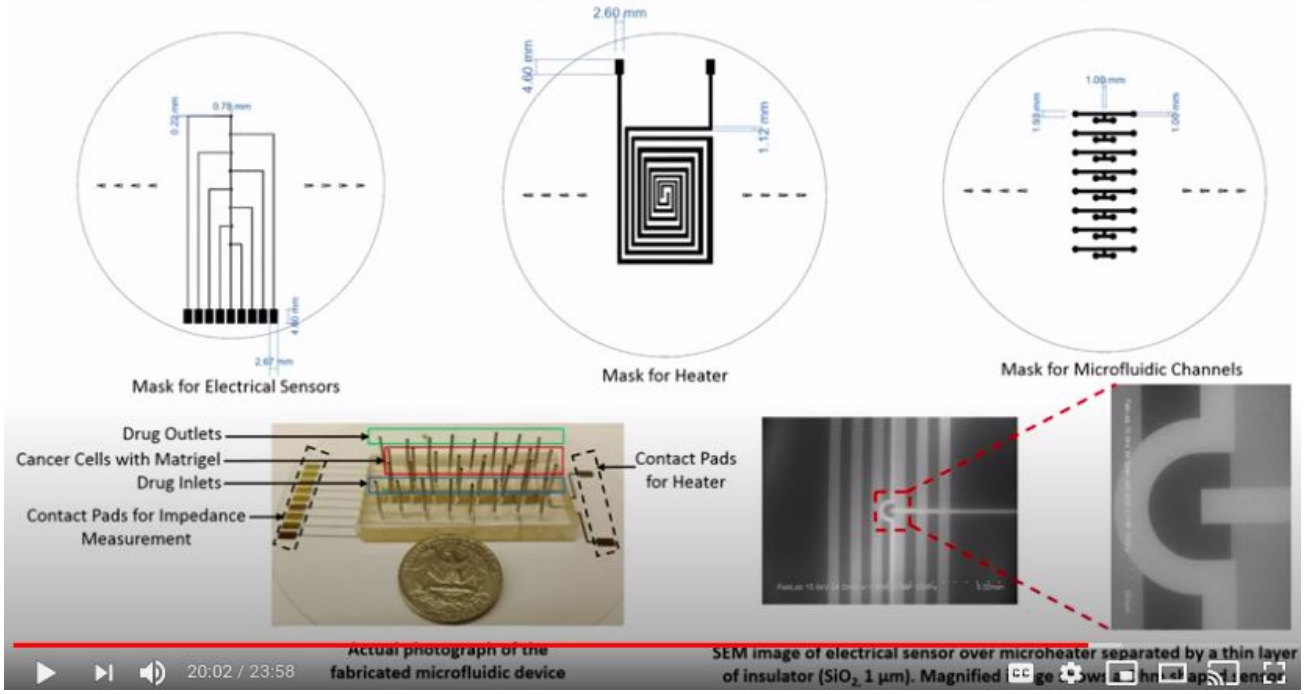
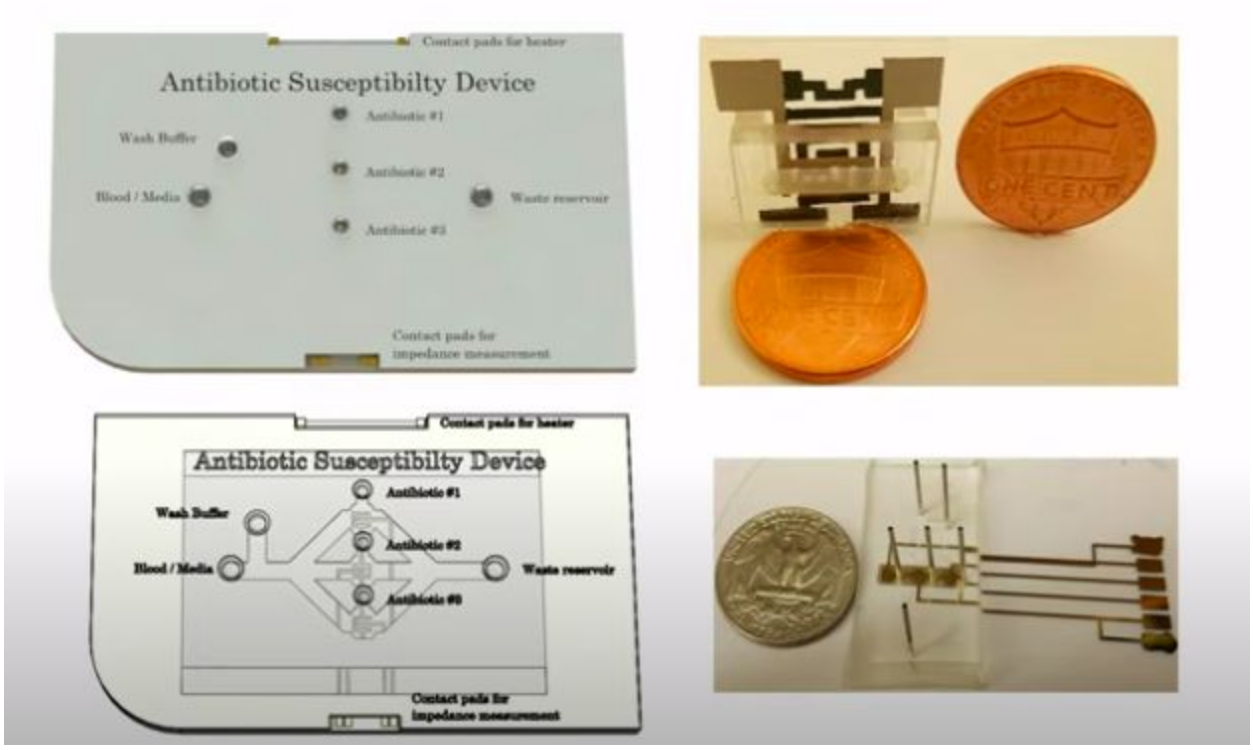
Microfluidic Chip for Rapid Drug Screening



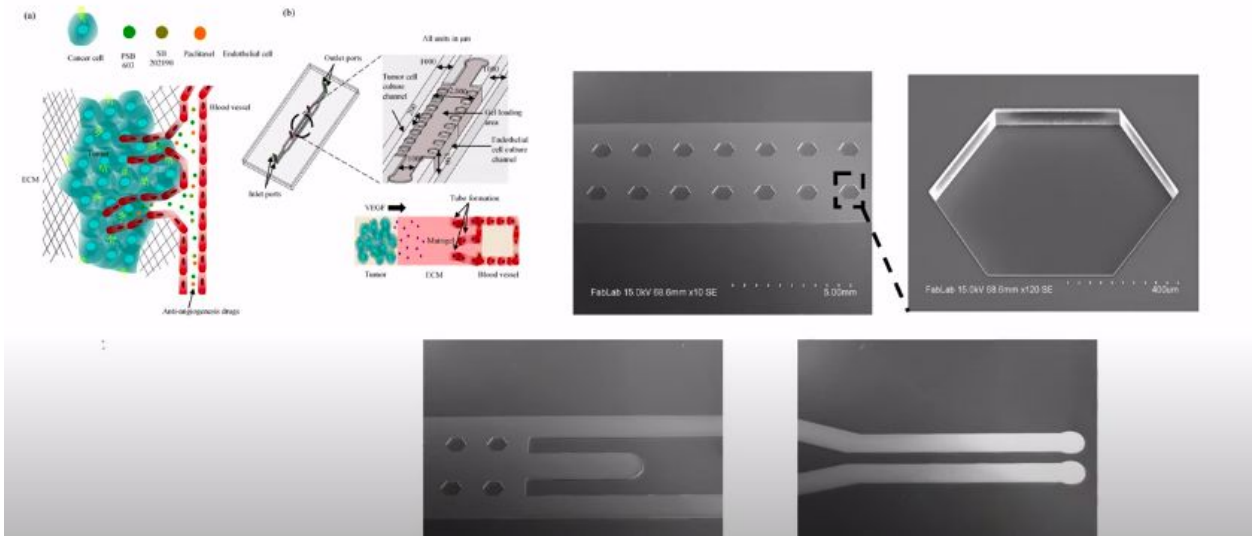
Microfluidic Device With Integrated Heater and Electrical Sensors for Drug Screening

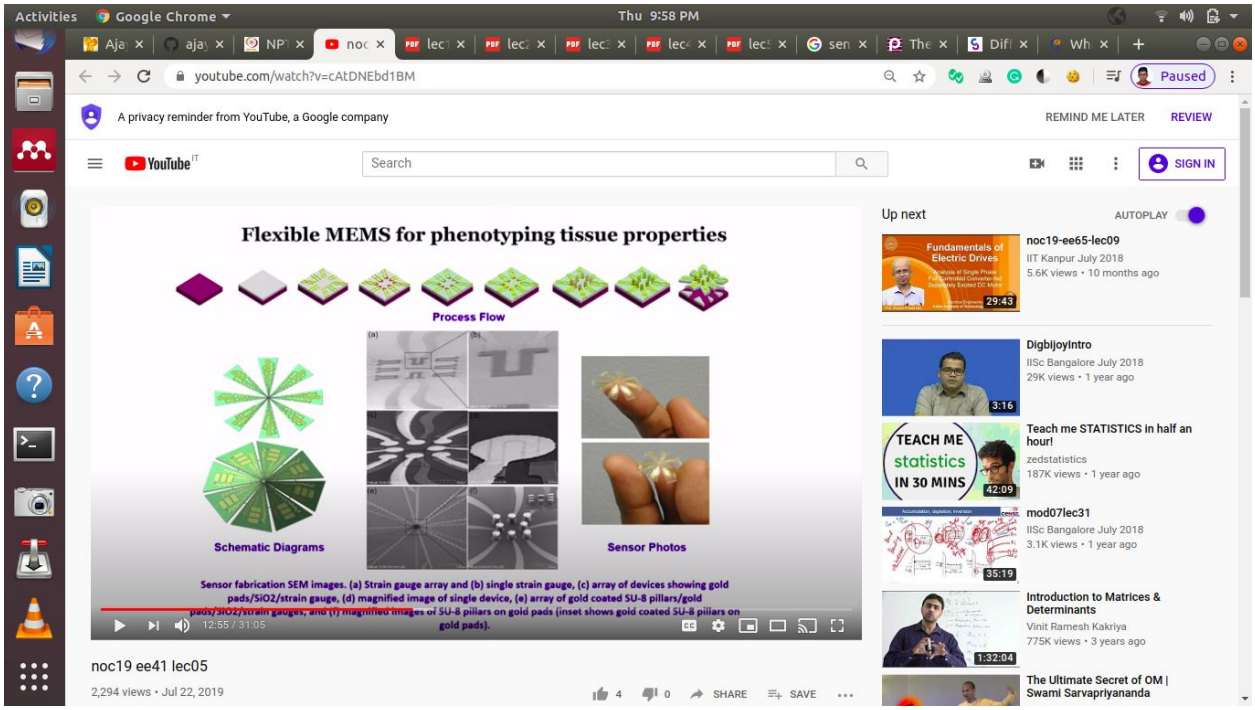


A Microfluidic Chip for rapid bacterial antibiotic susceptibility testing



Microfluidic Chip for Evaluating Drug Efficacy





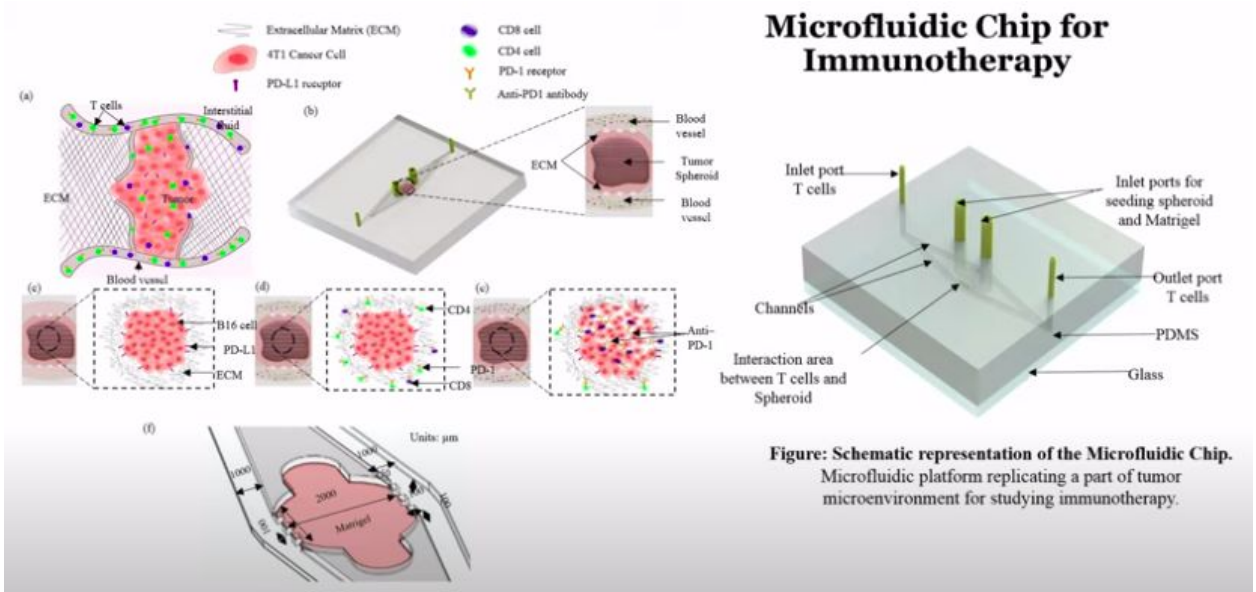
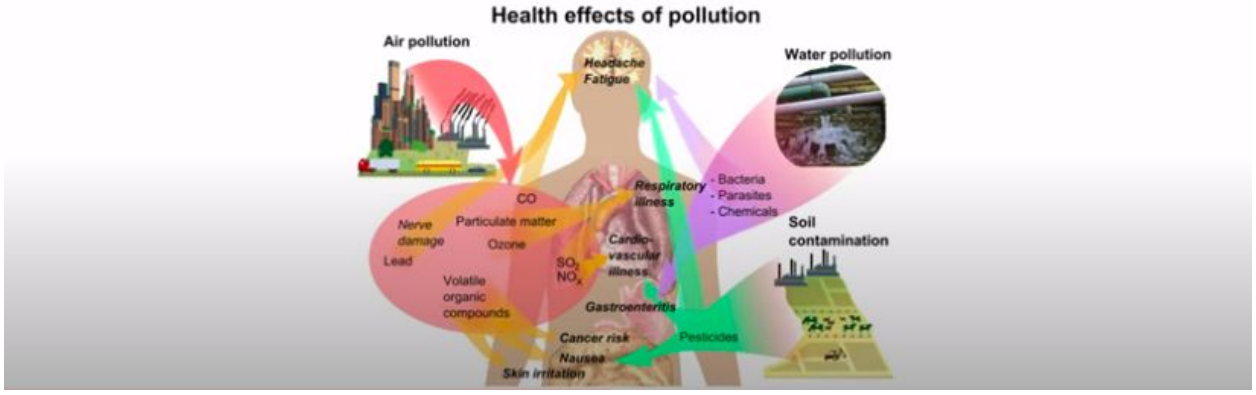


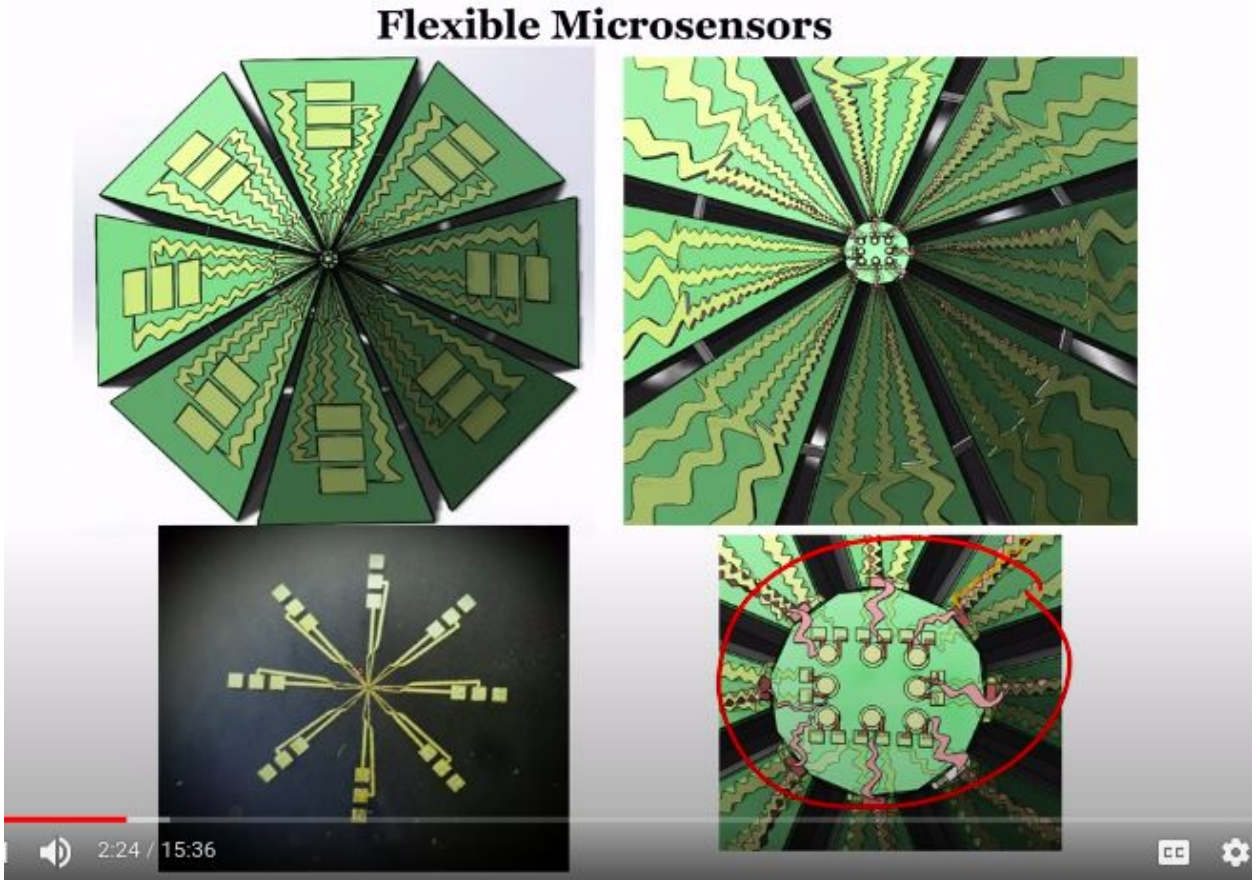
Figure: Schematic representation of the Microfluidic Chip. Microfluidic platform replicating a part of tumor microenvironment for studying immunotherapy.

Figure: Schematic representation of the Experimental Design. (a) Tumor microenvironment and interaction of tumor with T cells, **(b)** Microfluidic platform replicating a part of tumor microenvironment for studying immunotherapy, **(c)** 4T1 (mouse breast cancer) cells spheroids embedded in the extracellular matrix (ECM) on the device. PDL-1 is shown in only few cells to avoid ambiguity in diagram. **(d)** After flowing T cells without immunomodulator for 24 h in the neighbouring

Novel Techniques for MEMS-Based VOC Sensors Using Nanostructured Metal Oxides: Developing a non-invasive technique for Detecting Diabetes

- VOCs are found in individual's breath in hundreds of ppm range and only a small number of VOCs are common to everyone. The concentration of VOCs exhaled by patients for different disease is different than that in healthy person.
- VOC sensing is important in environment monitoring and medical diagnosis.





Overview

- Quick look at some common MEMS actuators
  - Piezoelectric
  - Thermal
  - Magnetic
- Next:
  - Electrostatic actuators
  - Actuators and mechanism
  - Beams

MEMS Actuation Options

Piezoelectric
Thermal
Magnetic
Electrostatic

Dynamics
  Beam bending
  Damping

7:07 / 15:36